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Variational principles of guiding centre motion

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An elementary but rigorous derivation is given for a variational principle for guiding centre motion. The equations of motion resulting from the variational principle (the drift equations) possess exact conservation laws for phase volume, energy (for time-independent systems), and angular momentum (for azimuthally symmetric systems). The results of carrying the variational principle to higher order in the adiabatic parameter are displayed. The behaviour of guiding centre motion in azimuthally symmetric fields is discussed, and the role of angular momentum is clarified. The application of variational principles in the derivation and solution of gyrokinetic equations is discussed.

1. Introduction

The usefulness of variational principles is widely appreciated in plasma physics and in physics in general. In this paper we report on a variational principle for guiding centre motion. First we display the result and discuss it, we indicate how it was derived, then discuss the role of angular momentum in guiding centre motion for azimuthally symmetric systems, and finally show how variational principles may be applied to gyrokinetic equations.

We write our variational principle in the general form

$$\delta \int L\, dt = 0, \tag{1}$$

where the Lagrangian L is given by

$$L = \frac{1}{\epsilon}\mathbf{A}^* . \dot{\mathbf{X}} + \epsilon\mu\dot{\Theta} - H. \tag{2}$$

Our notation is as follows. We choose units such that $e = m = c = 1$, and we let ϵ stand for the adiabatic ordering parameter. Physically, ϵ represents the ratio of gyroradius to scale length; but mathematically, the ϵ is really only a reminder of the order of various terms. Physical formulae result by setting $\epsilon = 1$ and restoring physical factors of e, m and c. We let $\mathbf{X}$ represent the guiding centre position, μ the magnetic moment, and Θ the gyrophase. H is the Hamiltonian, given by

$$H = \Phi + \mu B + \tfrac{1}{2}U^2, \tag{3}$$

in which Φ is the electric potential and U is the parallel velocity of the guiding centre. The vector $\mathbf{A}^*$, which we call the 'modified vector potential', is defined by

$$\mathbf{A}^* = \mathbf{A} + \epsilon U\hat{b}, \tag{4}$$

in which $\hat{b} = \mathbf{B}/B$. The vector $\mathbf{A}^*$ plays a prominent role in a Hamiltonian theory of guiding centre motion; it was first discovered by Morozov & Solov'ev (1966). All quantities in (2)–(4) are evaluated at the guiding centre position $\mathbf{X}$.

We assume that the $\mathbf{E} \times \mathbf{B}$ drifts are of the same order in ϵ as the ∇B and curvature drifts. This is accomplished by setting $\mathbf{E}_{\rm ph} = \epsilon \mathbf{E}$, where $\mathbf{E}_{\rm ph}$ is the physical electric field, and $\mathbf{E}$ is the symbol used in our analysis here. (It is convenient to use, as much as possible, only symbols which are $O(1)$, so that the order of any term is given by the factor ϵ^n which explicitly appears. Since we are assuming $\mathbf{E}_{\rm ph} = O(\epsilon)$, we eliminate $\mathbf{E}_{\rm ph}$ in favour of $\mathbf{E}$, which is $O(1)$.) Similarly, we set $\Phi_{\rm ph} = \epsilon \Phi$. We also allow the fields $\mathbf{E}$ and $\mathbf{B}$ to be slow functions of time, i.e. functions of $\tau = \epsilon t$. Thus, $\mathbf{E}$ and $\mathbf{B}$ are allowed to vary on the drift time-scale. It is only slightly more difficult to analyse the case in which $(\mathbf{E}_{\rm ph})_\perp = O(1)$ and $\partial/\partial t = O(1)$. However, the case we are presenting is the most common in practice, even though it is not the most general.

It is instructive to obtain the equations of motion from the variational principle of (1)–(2), before deriving (2) itself. We regard the Lagrangian of (2) as a function of the guiding centre variables $(\mathbf{X}, U, \Theta, \mu)$ and their time derivatives $(\dot{\mathbf{X}}, \dot{U}, \dot{\Theta}, \dot{\mu})$, in which the derivatives $\dot{U}$ and $\dot{\mu}$ happen not to appear. A convenient notation is to let $\mathbf{Z}$ or Z_i, $i = 1, \ldots, 6$, represent the six guiding centre variables. Then the Euler–Lagrange equations are

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{Z}_i}\right) - \frac{\partial L}{\partial Z_i} = 0. \tag{5}$$

Taking $Z_i = U, \Theta, \mu$, we find, first, $U = \hat{b} \,.\, \dot{\mathbf{X}}$, showing that U is indeed the parallel velocity of the guiding centre; secondly, $\dot{\mu} = 0$, showing that μ is a (formal) constant of motion; and thirdly, $\dot{\Theta} = B/\epsilon$, showing the rapid evolution of the gyrophase. Next, taking $Z_i = X_i$, $i = 1, 2, 3$, we find

$$\mathbf{E}^* + \frac{1}{\epsilon} \dot{\mathbf{X}} \times \mathbf{B}^* = \dot{U} \hat{b} + \mu \nabla B, \tag{6}$$

where the modified fields $\mathbf{E}^*$ and $\mathbf{B}^*$ are defined by

$$\begin{aligned} \mathbf{B}^* &= \nabla \times \mathbf{A}^* = \mathbf{B} + \epsilon U \nabla \times \hat{b}, \\ \mathbf{E}^* &= -\frac{\partial \mathbf{A}^*}{\partial \tau} - \nabla \Phi = \mathbf{E} - \epsilon U \frac{\partial \hat{b}}{\partial \tau}. \end{aligned} \tag{7}$$

Equation (6) can be solved for $\dot{\mathbf{X}}$ by crossing with $\hat{b}$ and by using $U = \hat{b} \,.\, \dot{\mathbf{X}}$. This gives

$$\dot{\mathbf{X}} = \frac{1}{B_\parallel^*} [U \mathbf{B}^* + \epsilon \hat{b} \times (\mu \nabla B - \mathbf{E}^*)]. \tag{8}$$

Likewise, dotting (6) with $\mathbf{B}^*$ gives

$$\dot{U} = \frac{-1}{B_\parallel^*} \mathbf{B}^* \,.\, (\mu \nabla B - \mathbf{E}^*), \tag{9}$$

which is the equation for the parallel acceleration. In (8)–(9), we have

$$B_\parallel^* = B + \epsilon U (\hat{b} \,.\, \nabla \times \hat{b}).$$

Equations (8)–(9) are the drift equations, and they are valid to the same order in ϵ as those given by Northrop (1963). To show the equivalence to Northrop's equations, one must express the modified fields $\mathbf{E}^*$ and $\mathbf{B}^*$ in terms of the real fields $\mathbf{E}$ and $\mathbf{B}$, and expand the factors $1/B_\parallel^*$ in powers of ϵ. One finds, for example, that the curvature drift is hidden in the term $U\mathbf{B}^*/B_\parallel^*$ of (8). Later we will show how to extend these drift equations to one higher order in ϵ.

The results (8)–(9) have been obtained previously (Littlejohn 1981). However, the previous method for deriving them was much more difficult than the one presented here, and it did not use a variational principle.

The Lagrangian of (2) will be rigorously derived by an elementary method in § 2. However, this should not be regarded as just another derivation of the usual drift equations. In particular, the denominators $B_\parallel^*$ in (8)–(9) have not been derived by other methods, but they are essential to the Hamiltonian properties of guiding centre motion, including the existence of canonical q's and p's, the relation between symmetries and constants of motion, conservation of energy, and Liouville's theorem. Without the denominators $B_\parallel^*$, the drift equations will not conserve angular momentum exactly in azimuthally symmetric systems, or energy (in some cases) in time-independent systems.

One of the advantages of variational formulations is their covariance with respect to arbitrary co-ordinate transformations. Since (1) is a scalar equation, generalized co-ordinates may be used, and the Euler–Lagrange equations will automatically give the correct equations of motion.

Consider, for example, the co-ordinate system $\mathbf{Z} = (\mathbf{X}, K, \Theta, \mu)$, where the energy $K = \Phi + \mu B + \frac{1}{2}U^2$ is used as a co-ordinate instead of the parallel velocity U. Since $\dot{U}$ does not appear in (2), the Lagrangian is transformed to the new co-ordinates simply by setting $H = K$; it does not contain $\dot{K}$. However, in the new co-ordinates, U is not an independent variable, but is considered a function of $\mathbf{X}$, K, and μ, given by

$$U = \pm\,[2(K - \mu B - \Phi)]^{\frac{1}{2}}. \tag{10}$$

The co-ordinates $(\mathbf{X}, K, \Theta, \mu)$ are most useful for time-independent systems, for which K is a constant; for purposes of illustration we may work through the variational equations in this case. We find

$$\dot{\mathbf{X}} = \frac{1}{B_\parallel^*} U\mathbf{B}^*, \quad \dot{K} = 0, \tag{11}$$

in place of (8)–(9). We still define $\mathbf{B}^*$ as $\nabla \times \mathbf{A}^*$, but the operator ∇ with K held fixed is not the same as ∇ with U held fixed, so $\mathbf{B}^*$ here is not the same as it was in (8)–(9). When this difference is accounted for, (11) is seen to be the same as (8)–(9). Equations (11) are striking in their simplicity, in spite of the fact that the definition of $\mathbf{B}^*$ hides some complexity. These equations were discovered previously by Boozer (1980).

As another example, consider representing the guiding centre position $\mathbf{X}$ in terms of cylindrical co-ordinates (r, z, ϕ). Then we have

$$\mathbf{A}^* \cdot \dot{\mathbf{X}} = A_r^* \dot{r} + A_z^* \dot{z} + r A_\phi^* \dot{\phi}, \tag{12}$$

which can be substituted into (2), and the new equations of motion follow immediately. Hamada or flux co-ordinates could be treated similarly.

Variational principles yield a simple relation between symmetries and constants of the motion (Noether's theorem). For example, the constancy of μ derives from the fact that $\partial L/\partial\Theta = 0$. Similarly, if a system possesses azimuthal symmetry, so that $\partial L/\partial\phi = 0$, then from (12) it is clear that rA_ϕ^* is constant. There has been some question in the literature in the past as to exactly how the 'angular momentum of the guiding centre' should be defined, or even whether an axially symmetric system possesses an exact constant of the guiding centre motion, analogous to the angular momentum of the particle. Here the angular momentum of the guiding centre appears unambiguously as $p_\phi = \partial L/\partial\dot{\phi} = rA_\phi^*/\epsilon$, and it is exactly conserved by the equations (8)–(9). We shall say more about angular momentum in § 5.

Our variational principle is not Hamilton's principle in the ordinary sense in mechanics, because we are independently varying all six phase space co-ordinates, and not just the three configuration space co-ordinates. In addition, not all of the variations are forced to vanish at the end points. For these reasons, we prefer to think of our variational principle as being a Hamiltonian construction (or at least a hybrid Hamiltonian–Lagrangian), rather than a true Lagrangian construction. In § 4 we establish the connexion between our variational principle and ordinary Hamiltonian mechanics, expressed in terms of p's and q's.

Our Lagrangian principle can be generalized to relativistic guiding centre motion, and it can also be used to analyse the longitudinal invariant and the bounce-averaged motion (Littlejohn 1982*b*). These generalizations will be described elsewhere.

A variational formulation of guiding centre motion has been given previously by Taylor (1964). Taylor's Lagrangian is evidently based on the analogy between the magnetic moment of gyration of a classical guiding centre and the intrinsic magnetic moment of a quantum mechanical particle, for he includes a term $-\mu B$ in his Lagrangian for the guiding centre. The approach we take in this paper differs in several respects. First, we begin with the true, exact Lagrangian of the particle, and derive the guiding centre Lagrangian and Hamiltonian from it. That is, our derivation is completely rigorous. Secondly, our transformation or averaging procedure is systematic, and can be carried to any desired order in the adiabatic parameter ϵ. And thirdly, we use rectangular (rather than field line) co-ordinates throughout, so that we avoid having to deal with the metric tensor g_{ij} for the field line co-ordinate system.

2. An elementary derivation of the guiding centre Lagrangian

The basic idea of (2) is that L is the gyroaverage of the Lagrangian of the particle, the latter being expressed as a function of the canonical variables $\mathbf{q}$ and $\mathbf{p}$. Without yet being specific as to the forces which act on the particle, we can write the particle Lagrangian as

$$L(\mathbf{q}, \mathbf{p}, \dot{\mathbf{q}}, t) = \mathbf{p} \cdot \dot{\mathbf{q}} - h_{\text{can}}(\mathbf{q}, \mathbf{p}, t) \tag{13}$$

where h_{can} is the usual Hamiltonian in terms of q's and p's. Hamilton's equations follow from this Lagrangian by varying $\mathbf{q}$ and $\mathbf{p}$ independently, subject to the

condition $\delta\mathbf{q}.\mathbf{p} = 0$ at the end points. This variational principle is discussed at greater length by Goldstein (1980, p. 362), Landau and Lifshitz (1969, p. 140), and Arnold (1978, p. 243).

Let us now write $\mathbf{z}$ or z_i, $i = 1, \ldots, 6$, for some set of six independent functions of $(\mathbf{q}, \mathbf{p}, t)$. The z's represent an arbitrary co-ordinate system on phase space. Substituting $\mathbf{z} = \mathbf{z}(\mathbf{q}, \mathbf{p}, t)$ or the inverse $\mathbf{q} = \mathbf{q}(\mathbf{z}, t)$, $\mathbf{p} = \mathbf{p}(\mathbf{z}, t)$ into (13) and using the chain rule gives

$$L(\mathbf{z}, \dot{\mathbf{z}}, t) = \sum_{i=1}^{6} \gamma_i \dot{z}_i - h, \tag{14}$$

where γ_i and h are functions of $\mathbf{z}$ and t, given by

$$\gamma_i(\mathbf{z}, t) = \mathbf{p}.\frac{\partial \mathbf{q}}{\partial z_i}, \tag{15}$$

$$h(\mathbf{z}, t) = h_{\text{can}} - \mathbf{p}.\frac{\partial \mathbf{q}}{\partial t}. \tag{16}$$

In general, all of the $\dot{\mathbf{z}}$'s will appear in (14).

The form of the Lagrangian shown in (14) is completely general. Only the functions $\gamma_i(\mathbf{z}, t)$ and $h(\mathbf{z}, t)$ will change on passing from one co-ordinate system to another, and the correct equations of motion will always follow by varying the integral of $L(\mathbf{z}, \dot{\mathbf{z}}, t)\, dt$. One subtle point is that the variations of all six phase space co-ordinates $\mathbf{z}$ are to be considered independent during the variational process, even if a functional dependence should result through the equations of motion. For example, in the canonical co-ordinates $\mathbf{z} = (\mathbf{q}, \mathbf{p})$, the momenta $\mathbf{p}$ are not considered to be functions of $\mathbf{q}$ and $\dot{\mathbf{q}}$ during the variation, even though such a relation will follow from the equations of motion.

In the case of a particle in an electromagnetic field, the canonical Hamiltonian is

$$h_{\text{can}}(\mathbf{q}, \mathbf{p}, t) = \frac{1}{2}\left[\mathbf{p} - \frac{1}{\epsilon}\mathbf{A}(\mathbf{q}, \epsilon t)\right]^2 + \Phi(\mathbf{q}, \epsilon t), \tag{17}$$

where ϵ has been appropriately introduced. Let us now write $\mathbf{z} = (\mathbf{x}, \mathbf{v})$, where $\mathbf{x}$ is the particle position and $\mathbf{v}$ is the velocity; these are related to the canonical variables by $\mathbf{x} = \mathbf{q}$ and $\mathbf{v} = \mathbf{p} - \mathbf{A}(\mathbf{q}, \epsilon t)/\epsilon$. Then we find, corresponding to (14),

$$L = \left[\frac{1}{\epsilon}\mathbf{A}(\mathbf{x}, \epsilon t) + \mathbf{v}\right].\dot{\mathbf{x}} - [\Phi(\mathbf{x}, \epsilon t) + \tfrac{1}{2}v^2]. \tag{18}$$

Note that $\mathbf{v}$ is considered independent of $\dot{\mathbf{x}}$, even though $\mathbf{v} = \dot{\mathbf{x}}$ follows from the equations of motion. This is because the variational principle selects the physical motion, out of all conceivable motions, as the one to make the action integral stationary. Although $\mathbf{v} = \dot{\mathbf{x}}$ on the physical motion, this will not be true along all conceivable paths through $(\mathbf{x}, \mathbf{v})$ space.

It is easy to average (18) to lowest order. Let us write $u = v_{\parallel}$ and $w = v_{\perp}$, and decompose the particle velocity according to

$$\mathbf{v} = u\hat{b} + w\hat{c}, \tag{19}$$

where $\hat{c}$ is a perpendicular unit vector. At lowest order, u, $\hat{b}$, and w are purely averaged, but $\hat{c}$ is purely oscillatory. Thus, we have $\mathbf{v} = u\hat{b} + \text{osc}$. Similarly, let

us write $\mathbf{x} = \mathbf{X} + \epsilon(\text{osc})$, where $\mathbf{X}$ is the guiding centre position. Substituting these relations into (18) and neglecting oscillatory terms, we find the gyroaverage of (18) up to $O(\epsilon^0)$:

$$L = \left[\frac{1}{\epsilon}\mathbf{A}(\mathbf{X}, \epsilon t) + u\hat{b}\right].\dot{\mathbf{X}} - [\Phi(\mathbf{X}, \epsilon t) + \tfrac{1}{2}w^2 + \tfrac{1}{2}u^2] + O(\epsilon). \tag{20}$$

This result already bears a strong resemblance to (2), showing in particular that the modified vector potential $\mathbf{A}^*$ is nothing more than the gyroaverage of the canonical momentum $\mathbf{p}$. However, it differs in several respects from (2), for it is expressed in terms of w instead of $\mu = w^2/2B$, it uses u instead of U, and the term $\epsilon\mu\dot{\Theta}$ is missing. These differences will be clarified as we proceed.

We now define the instantaneous gyrophase θ implicitly by the relations

$$\left.\begin{aligned} \hat{a} &= \cos\theta\,\hat{e}_1 - \sin\theta\,\hat{e}_2, \\ \hat{c} &= -\sin\theta\,\hat{e}_1 - \cos\theta\,\hat{e}_2, \end{aligned}\right\} \tag{21}$$

where $\hat{c}$ is defined by (19), where $\hat{a} = \hat{b} \times \hat{c}$, and $\hat{e}_1$ and $\hat{e}_2$ are some arbitrary pair of perpendicular unit vectors, satisfying $\hat{e}_1 \times \hat{e}_2 = \hat{b}$. The vectors $\hat{b}, \hat{e}_1, \hat{e}_2$ are functions of $(\mathbf{x}, \epsilon t)$.

The $O(\epsilon)$ term of (20) is important, and it must be obtained by transforming (18) to guiding centre variables, instead of simply neglecting oscillatory terms. Let us return to (18) and transform from the particle position $\mathbf{x}$ to the guiding centre position $\mathbf{X}$, according to the relation

$$\mathbf{x} = \mathbf{X} + \epsilon w\hat{a}/B, \tag{22}$$

where the right-hand side is evaluated at $\mathbf{X}$. We now regard the variables $(\mathbf{X}, u, \theta, w)$ as the six phase space co-ordinates. Substituting into (18) and using the chain rule gives a power series in ϵ. The dominant, $O(\epsilon^{-1})$ term is $(1/\epsilon)\mathbf{A}.\dot{\mathbf{X}}$, just as in (20). The next term, at $O(\epsilon^0)$, is

$$\frac{w}{B}\hat{a}.\nabla\mathbf{A}.\dot{\mathbf{X}} + (w\hat{c} + u\hat{b}).\dot{\mathbf{X}} + \frac{d}{dt}\left(\frac{w\hat{a}}{B}\right).\mathbf{A}. \tag{23}$$

Before proceeding with this, we note a general rule for transforming Lagrangians. Let $L = L(\mathbf{z}, \dot{\mathbf{z}}, t)$ be a Lagrangian of the form of (14), and let $S = S(\mathbf{z}, t)$ be an arbitrary scalar. Then the equations of motion are invariant under the substitution $L \to L + dS/dt$, i.e. under

$$\gamma_i \to \gamma_i + \frac{\partial S}{\partial z_i}, \quad h \to h - \frac{\partial S}{\partial t}. \tag{24}$$

We will call such a transformation a gauge transformation in phase space, to distinguish it from an ordinary, electromagnetic gauge transformation, which it resembles. The two kinds of transformations on Lagrangians of the form of (14), changes of co-ordinates and gauge transformations, allow us to average the particle Lagrangian, (18), to any order we wish, and to obtain the guiding centre Lagrangian.

In particular, by taking

$$S = -(w/B)\hat{a}.\mathbf{A}, \tag{25}$$

evaluated at $(\mathbf{X}, \epsilon t)$, the expression (23) becomes clearer. The motivation is to get rid of the (ordinary, electromagnetic) gauge-dependent terms, i.e. the terms in $\mathbf{A}$. When dS/dt is expanded, the result cancels several terms of (23), leaving finally only $u\hat{b}.\dot{\mathbf{X}}$, just as in (20).

Continuing, at $O(\epsilon)$ we obtain the term

$$\epsilon\left[(\ldots).\dot{\mathbf{X}} + \left(\frac{w^2}{B^2}\hat{a}.\nabla\mathbf{A}.\hat{c} + \frac{w^2}{B}\right)\dot{\theta} + \left(\frac{w}{B^2}\hat{a}.\nabla\mathbf{A}.\hat{a}\right)\dot{w}\right], \tag{26}$$

where we omit the somewhat complicated expression multiplying $\dot{\mathbf{X}}$, as it is not needed here. (It would be required for the second-order drifts.) To this we add dS/dt, where S is now given by

$$S = -\epsilon\frac{w^2}{2B^2}\hat{a}.\nabla\mathbf{A}.\hat{a}, \tag{27}$$

in order to cancel the (ordinary) gauge-dependent terms. This combines with the expression (26) to give simply

$$\epsilon\left[(\ldots).\dot{\mathbf{X}} + \frac{w^2}{2B}\dot{\theta}\right]. \tag{28}$$

Altogether, we have shown that the substitution (22) gives

$$L = \left[\frac{1}{\epsilon}\mathbf{A} + u\hat{b} + O(\epsilon)\right].\dot{\mathbf{X}} + \left[\epsilon\frac{w^2}{2B} + O(\epsilon^2)\right]\dot{\theta} + [O(\epsilon^2)]\,\dot{u} + [O(\epsilon^2)]\,\dot{w} - [\Phi + \tfrac{1}{2}w^2 + \tfrac{1}{2}u^2 + O(\epsilon)], \tag{29}$$

in which the order of all neglected terms is carefully shown. Note that everything is now evaluated at $\mathbf{X}$.

This Lagrangian is independent of the gyrophase θ, and so by Noether's theorem the quantity $\partial L/\partial\dot{\theta} = \epsilon w^2/2B = \epsilon\mu$ is a constant. This encourages us to eliminate w in favour of μ, which brings (29) closer to the form of (2). Note that this substitution produces the term $\epsilon\mu\dot{\theta}$. The final differences between (2) and (29) are the use of the variables (U, Θ) in the former, and (u, θ) in the latter. The capitalization of the variables U and Θ is merely our way of indicating the averaged, guiding centre variables, in contrast to the particle variables u and θ. But since (u, θ) and (U, Θ) are equal at lowest order in ϵ, (29) is valid as it stands with either (u, θ) or (U, Θ). Thus we arrive at (2).

The derivation of the guiding centre Lagrangian, (2), from the particle Lagrangian, (18), can be extended to arbitrarily high order. We have actually carried out this calculation, to one higher order in ϵ than shown in (29). Instead of proceeding in an *ad hoc* and unsystematic manner, as shown here in § 2 for the lower-order calculations, we have applied a variant of the Lie transform technique, which is a particularly elegant way of carrying out systematic calculations. Our use of Lie transforms is based on the Hamiltonian methods of Dragt & Finn (1976), which are reviewed by Cary (1981), but they are generalized for application to Lagrangians of the form of (14), as described by Cary & Littlejohn (1982) and Littlejohn (1982*a*). We believe that this method is the best one now available for deriving higher-order terms in the guiding centre expansion.

However, for most practical purposes it is probably more important to know that a systematic procedure exists than it is actually to carry it out. Therefore, instead of presenting our calculations in detail, we merely display the results. Some of the qualitative features of this calculation are of importance, as will be indicated. We prefer to communicate privately with interested readers concerning the details of this calculation.

3. Results of the higher-order calculation

The guiding centre Lagrangian is expressed in terms of the guiding centre variables $(\mathbf{X}, U, \Theta, W)$, which are definite functions of the particle variables $(\mathbf{x}, u, \theta, w)$. The transformation connecting the two is a generalization of (22), and is given by

$$\mathbf{X} = \mathbf{x} - \epsilon\frac{w\hat{a}}{B} + \epsilon^2\left(\left[-\frac{w^2}{2B^3}(\hat{a}.\nabla B) + \frac{wu}{B^2}(\hat{b}.\nabla\times\hat{b})\right]\hat{a} + \left\{-\frac{w^2}{8B^2}[5(aba)-(cbc)] + \frac{2uw}{B^2}(bbc)\right\}\hat{b}\right) + O(\epsilon^3), \quad (30a)$$

$$U = u + \epsilon\left[-\frac{w^2}{2B}(abc) - \frac{uw}{B}(bba)\right] + O(\epsilon^2), \quad (30b)$$

$$\Theta = \theta + \epsilon\left\{-\frac{w}{B}(\hat{a}.R) + \frac{w}{B^2}(\hat{c}.\nabla B) - \frac{1}{wB}(\hat{c}.\mathbf{E}) - \frac{u}{4B}[(aba)-(cbc)] + \frac{u^2}{wB}(bbc)\right\} + O(\epsilon^2), \quad (30c)$$

$$W = w + \epsilon\left\{-\frac{1}{B}(\hat{a}.\mathbf{E}) + \frac{uw}{4B}[3(abc)-(cba)] + \frac{u^2}{B}(bba)\right\} + O(\epsilon^2), \quad (30d)$$

where the symbols (abc), (bba), etc. are abbreviations for $(\hat{a}.\nabla\hat{b}).\hat{c}$, $(\hat{b}.\nabla\hat{b}).\hat{a}$, etc., and where $\mathbf{R} = (\nabla\hat{e}_1).\hat{e}_2$.

The guiding centre variables are free of rapid oscillations (including Θ itself). As a function of the particle variables, they are only known to the order shown, but there exist algorithms for carrying them to arbitrary order. The requirement that the variables be free of rapid oscillations does not uniquely determine the higher-order terms, but it can be shown that there is one choice for these higher-order terms which causes the Lagrangian to simplify. We will refer to this choice as the 'standard guiding centre variables', and we will presently say more about them.

In terms of standard guiding centre variables, the magnetic moment is given by $\mu = W^2/2B(\mathbf{X}, \tau)$ to all orders. In terms of particle variables, this is

$$\mu = \frac{w^2}{2B} + \epsilon\left\{-\frac{w}{B^2}(\hat{a}.\mathbf{E}) + \frac{w^2u}{4B^2}[3(abc)-(cba)] + \frac{wu^2}{B^2}(bba) + \frac{w^3}{2B^3}(\hat{a}.\nabla B)\right\} + O(\epsilon^2). \quad (31)$$

This is in agreement with the original calculation of Kruskal (1965).

In terms of standard guiding centre variables, the guiding centre Lagrangian

has precisely the form shown already in (2), except that now the modified vector potential $\mathbf{A}^*$ is given by

$$\mathbf{A}^* = \mathbf{A} + \epsilon U \hat{b} - \epsilon^2 \mu \mathbf{R}, \tag{32}$$

where all higher-order terms vanish, and the new Hamiltonian is

$$H = (\Phi + \mu B + \tfrac{1}{2}U^2) + \epsilon[\tfrac{1}{2}\mu U(\hat{b}.\nabla \times \hat{b})] + O(\epsilon^2). \tag{33}$$

The terms multiplying $\dot{U}$ and $\dot{W}$ in the guiding centre Lagrangian vanish to all orders, and the term multiplying $\dot{\Theta}$ is $\epsilon W^2/2B = \epsilon\mu$ to all orders (cf. (29), where finite-order remainder terms are shown).

The higher-order terms in the drift equations are easily obtained by applying the Euler–Lagrange equations to this Lagrangian. The results are in agreement with the earlier calculation of Northrop & Rome (1978), but they include electric fields and they are expressed in terms of slightly different variables. Northrop & Rome achieved their higher-order calculation by applying a technique due to Kruskal, as explained by Northrop (1963, p. 36–40). The guiding centre variable U is found not to be exactly the parallel component of the guiding centre velocity $\hat{b}(\mathbf{X}, \tau).\dot{\mathbf{X}}$, but rather it has an $O(\epsilon)$ correction. This correction term has been given a clear physical interpretation by Northrop & Rome (1978).

The term involving $\mathbf{R} = (\nabla \hat{e}_1).\hat{e}_2$ in (32) appears peculiar at first, because it seems strange that $\mathbf{A}^*$ should depend on the choice of perpendicular unit vectors $\hat{e}_1$ and $\hat{e}_2$. Actually, the function of this term in $\mathbf{A}^*$ is to cause the entire Lagrangian of (2) to be invariant under a redefinition of $\hat{e}_1$ and $\hat{e}_2$. Such a redefinition is actually a gauge transformation of a certain kind, which we call a gyrogauge transformation. Without going into details, we can say that the role of the term $-\epsilon^2\mu\mathbf{R}$ in $\mathbf{A}^*$ is to cancel the gyrogauge dependence of the term $\epsilon\mu\dot{\Theta}$ in the Lagrangian (since Θ changes under a gyrogauge transformation). The result of using a gyrogauge invariant Lagrangian is that the equations of motion (the drift equations) are gyrogauge invariant to all orders.

The use of standard guiding centre variables does not yield any gain in accuracy, since these variables are only known to the order shown in (30). But their advantage is that they satisfy a number of exact relations among themselves, such as $\mu = W^2/2B$, avoiding the need to retain error estimates in various manipulations of these variables (such as in (29)). When standard guiding centre variables are used, all the error estimates of the Lagrangian are conveniently contained in the Hamiltonian of (33).

4. The connexion with ordinary Hamiltonian mechanics

In (14) we displayed the general form into which the phase space Lagrangian of any dynamical system (of three degrees of freedom) can be cast. The following discussion is quite general, and need not be restricted to particles or guiding centres. The co-ordinates z_i represent an arbitrary, possibly non-canonical, possibly time-dependent co-ordinate system on phase space. Here we shall establish the connexions between the formalism developed here and ordinary Hamiltonian mechanics, expressed in terms of q's and p's, as well as with our own previous researches on this subject (Littlejohn 1979, 1981).

Equation (14) was obtained from (13) by starting with canonical variables and by performing a possibly non-canonical co-ordinate transformation

$$\mathbf{z} = \mathbf{z}(\mathbf{q}, \mathbf{p}, t).$$

The statements (34)–(41) below can be verified by working backwards through this transformation to the known structure of Hamiltonian systems in terms of q's and p's (Hamilton's equations, Poisson brackets, etc.). The motivation for using the Lagrangians (13) and (14) is that Lagrangians are automatically covariant under arbitrary changes of co-ordinates.

It is straightforward to apply the Euler–Lagrange equations to the Lagrangian (14). One finds

$$\sum_{j=1}^{6} \omega_{ij} \dot{z}_j = \frac{\partial h}{\partial z_i} + \frac{\partial \gamma_i}{\partial t}, \tag{34}$$

where

$$\omega_{ij} = \frac{\partial \gamma_j}{\partial z_i} - \frac{\partial \gamma_i}{\partial z_j}. \tag{35}$$

The quantities ω_{ij} form an invertible matrix. The ω_{ij} are the Lagrange brackets (Goldstein 1980, p. 401) of the co-ordinates among themselves, denoted here by square brackets:

$$\omega_{ij} = [z_i, z_j] = \frac{\partial \mathbf{p}}{\partial z_i} \cdot \frac{\partial \mathbf{q}}{\partial z_j} - \frac{\partial \mathbf{p}}{\partial z_j} \cdot \frac{\partial \mathbf{q}}{\partial z_i}. \tag{36}$$

We define a matrix J_{ij} as the inverse of ω_{ij}:

$$\sum_{k=1}^{6} J_{ik} \omega_{kj} = \delta_{ij}. \tag{37}$$

The matrix J_{ij} can be used to solve (34) explicitly for the equations of motion,

$$\dot{z}_i = \sum_{j=1}^{6} J_{ij} \left(\frac{\partial h}{\partial z_j} + \frac{\partial \gamma_j}{\partial t} \right). \tag{38}$$

These are really Hamilton's equations, slightly disguised. The quantities J_{ij} are the Poisson brackets of the co-ordinates among themselves, here denoted by curly brackets:

$$J_{ij} = \{z_i, z_j\} = \frac{\partial z_i}{\partial \mathbf{q}} \cdot \frac{\partial z_j}{\partial \mathbf{p}} - \frac{\partial z_i}{\partial \mathbf{p}} \cdot \frac{\partial z_j}{\partial \mathbf{q}}. \tag{39}$$

If D is the Jacobian $\partial(\mathbf{q}, \mathbf{p})/\partial z$, so that $d^3\mathbf{q}\, d^3\mathbf{p} = D d^6\mathbf{z}$, then

$$\det(\omega_{ij}) = D^2. \tag{40}$$

Thus D can be calculated directly in the co-ordinates $\mathbf{z}$, and one need not have recourse to the original q's and p's.

Liouville's theorem can be expressed in terms of D:

$$\frac{\partial D}{\partial t} + \sum_{i=1}^{6} \frac{\partial}{\partial z_i} (D \dot{z}_i) = 0, \tag{41}$$

where $\dot{z}_i$ is given by (38). This is the same as the usual Liouville's theorem, but it is expressed in the co-ordinates $\mathbf{z}$. It can be proved directly by using (38) and the formula for the derivative of a determinant.

When these relationships are applied to the guiding centre variables $(\mathbf{X}, U, \Theta, \mu)$, we find the following. First, the Lagrange brackets are

$$[X_i, X_j] = \frac{1}{\epsilon} B^*_{ij}, \quad [\mathbf{X}, U] = -\hat{b}, \quad [\mathbf{X}, \mu] = \epsilon \mathbf{R}. \tag{42}$$

In these equations, the tensor B^*_{ij} is dual to the vector $\mathbf{B}^*$, i.e. $B^*_{ij} = \epsilon_{ijk} B^*_k$, where ϵ_{ijk} is the Levi-Civita symbol. $\mathbf{B}^*$ in turn is defined by $\mathbf{B}^* = \nabla \times \mathbf{A}^*$, where $\mathbf{A}^*$ is given by (32). The Lagrange brackets not listed are zero. Next, the Poisson brackets are given by

$$\left.\begin{aligned} \{X_i, X_j\} = -\frac{\epsilon}{B^*_\parallel} b_{ij}, \quad \{U, \Theta\} = -\frac{1}{B^*_\parallel} \mathbf{B}^* . \mathbf{R}, \\ \{\mathbf{X}, U\} = \frac{1}{B^*_\parallel} \mathbf{B}^*, \quad \{\mathbf{X}, \Theta\} = \frac{\epsilon}{B^*_\parallel} \hat{b} \times \mathbf{R}, \quad \{\Theta, \mu\} = \frac{1}{\epsilon}, \end{aligned}\right\} \tag{43}$$

where b_{ij} is dual to $\hat{b}$. The Poisson brackets not listed are zero. Finally, the Jacobian D is given by

$$D = B^*_\parallel. \tag{44}$$

If standard guiding centre variables are used, then the relations (42)–(44) are exact. Results quite similar to these have been derived by us previously, by much more complicated means (Littlejohn 1979, 1981).

Our phase space Lagrangian $L = L(\mathbf{z}, \dot{\mathbf{z}}, t)$ of (14) is distinct from the more usual configuration space Lagrangian, $L = L(\mathbf{q}, \dot{\mathbf{q}}, t)$. Our phase space Lagrangian has twice as many independent variables as the configuration space Lagrangian, and it is always linear in the time derivatives. This linearity is central to our use of Lie transforms in averaging the Lagrangian. Other averaging procedures have been developed by Whitham (1974) for configuration space Lagrangians, and applied by him to partial differential equations.

5. Azimuthal symmetry and angular momentum

Let us elaborate on the discussion surrounding (12). In an azimuthally symmetric system, the components of $\mathbf{A}$ with respect to the cylindrical co-ordinates (r, ϕ, z) are independent of the azimuthal angle ϕ, as are the components of $\mathbf{B}$, $\hat{b}$, ∇B, etc. It is in this sense that we must understand a vector to be azimuthally symmetric.

It is sometimes more convenient to work with the covariant components of a vector than the standard components obtained by orthogonal projections. If $\mathbf{V}$ is an arbitrary vector, then the covariant components (V_r, V_ϕ, V_z) are related to the standard ones by $V_r = \hat{r} . \mathbf{V}$, $V_\phi = r\hat{\phi} . \mathbf{V}$, and $V_z = \hat{z} . \mathbf{V}$. (Only the ϕ component is different, by a factor of r.) On the other hand, it is standard to define $\mathbf{p} = \partial L / \partial \dot{\mathbf{x}}$ and $p_\phi = \partial L / \partial \dot{\phi}$, so that in the case of p_ϕ, the standard form *is* the covariant form. We would like to point out that the formulae of this section reveal a greater symmetry when expressed exclusively in terms of the covariant components. Nevertheless, to avoid confusion we will stay with the standard notation.

As we pointed out previously, if the guiding centre Lagrangian is independent of the azimuthal angle ϕ, then $p_\phi = \partial L/\partial\dot{\phi}$ is a constant of the guiding centre motion. We will now show that this constant, the 'angular momentum of the guiding centre', is identical with the angular momentum of the particle to all orders.

The exact particle Lagrangian $\mathbf{p}.\dot{\mathbf{q}} - H$ can be transformed to cylindrical co-ordinates and written in the form

$$L = p_r\dot{r} + p_\phi\dot{\phi} + p_z\dot{z} - H. \tag{45}$$

Of course, the (r, ϕ, z) co-ordinates of the particle are not the same as those of the guiding centre, but the transformation connecting them has the form

$$r = r_{gc} + \ldots, \quad \phi = \phi_{gc} + \ldots, \quad z = z_{gc} + \ldots . \tag{46}$$

The omitted terms are functions of r_{gc} and z_{gc}, but, if the system is azimuthally symmetric, they are independent of ϕ_{gc}. This is because the transformation is built up out of the fields $\mathbf{E}$ and $\mathbf{B}$ and their derivatives, which are azimuthally symmetric. Therefore we can substitute (46) into the particle Lagrangian to obtain the guiding centre Lagrangian, and we easily see that the coefficient of $\dot{\phi}$ in the particle Lagrangian is identical with the coefficient of $\dot{\phi}_{gc}$ in the guiding centre Lagrangian. (If gauge transformations in phase space are performed, according to (24), they will not affect this conclusion, as long as the scalars S are also azimuthally symmetric.) Of course, the two expressions, although numerically equal, are expressed in terms of different variables.

When standard guiding centre variables are used, p_ϕ of the guiding centre can be expressed in closed form in terms of its variables. Specifically, we have $p_\phi = (1/\epsilon) r A^*_\phi$ exactly, with $\mathbf{A}^*$ given by (32). Of course, the transformation connecting particle variables with guiding centre variables will only be known to some finite order. But even if some approximation to the guiding centre Lagrangian is used, such as (2), then $\partial L/\partial\dot{\phi}$ will still be an exact constant of motion of the equations generated by the approximate Lagrangian.

The second-order term in the expression for $p_\phi = (1/\epsilon) r A^*_\phi$ is puzzling at first, because it contains the ϕ component of the vector $\mathbf{R} = (\nabla\hat{e}_1).\hat{e}_2$, as shown by (32). The problem is that p_ϕ is a physical quantity, and must be gyrogauge invariant, whereas $\mathbf{R}$ is not gyrogauge invariant. But the problem is only apparent. If a system possesses azimuthal symmetry, then the vectors $\hat{e}_1$ and $\hat{e}_2$ can always be chosen to have azimuthal symmetry, and we shall insist on this. It then follows that the component R_ϕ is gyrogauge invariant, since

$$R_\phi = \hat{\phi}.\nabla\hat{e}_1.\hat{e}_2 = r^{-1}(\partial\hat{e}_1/\partial\phi).\hat{e}_2 = r^{-1}(\hat{z}\times\hat{e}_1).\hat{e}_2 = r^{-1}b_z.$$

Thus, the angular momentum in standard guiding centre co-ordinates is given by

$$p_\phi = \frac{1}{\epsilon} r A_\phi + r U b_\phi - \epsilon\mu b_z, \tag{47}$$

where r is the r co-ordinate of the guiding centre.

6. Gyrokinetic equations

At present there is a considerable amount of research activity with gyrokinetic equations. Derivations and manipulations of these equations often involve enormous amounts of algebra, and we feel that the use of Lagrangian and Hamiltonian techniques can simplify much of this.

A detailed discussion of gyrokinetic equations is out of place in this paper, which is devoted to single particle concepts. Here we will simply discuss some considerations of a general nature on the Vlasov equation, its transformation properties, and means of solution, and we will defer detailed derivations to a later paper.

There are two complementary ways to solve the Vlasov equation, one using the Vlasov equation itself, and the other using single particle orbits. The former point of view has been developed at great length in a monumental work by Hastie, Taylor & Haas (1967); here we favour the latter point of view, because it allows us to apply Hamiltonian methods. In the case of uniform magnetic fields, single-particle Hamiltonian methods have been used by Cary (1979) for studying the nonlinear evolution of wave packets; part of our aim here is to extend those methods to the case of non-uniform magnetic fields.

It is useful to distinguish two versions of the distribution function, one, which we denote by F, transforming as a pseudo-scalar, and the other, which we denote by f, transforming as a scalar. We agree to set $f = F$ in the $(\mathbf{q}, \mathbf{p})$ co-ordinates, so that under the transformation $(\mathbf{q}, \mathbf{p}) \to \mathbf{z}$, we have $f(\mathbf{z}) = f(\mathbf{q}, \mathbf{p})$ and

$$F(\mathbf{z}) = DF(\mathbf{q}, \mathbf{p}),$$

where D is given by (40). Thus, in an arbitrary co-ordinate system, we will have $F = Df$. (The more usual $(\mathbf{x}, \mathbf{v})$ co-ordinates are trivially related to the $(\mathbf{q}, \mathbf{p})$ co-ordinates by the constant Jacobian m^3, which is unity in the units chosen here. All canonical co-ordinate systems are related to one another by unity Jacobian, so that $f = F$ in all canonical co-ordinates.)

Much of the algebra required in using gyrokinetic equations comes from transforming various partial differential operators from one co-ordinate system to another. Sometimes a great deal of effort can be saved by calling on some general observations. For example, the Vlasov equation in the $(\mathbf{x}, \mathbf{v})$ co-ordinates can be written in the form

$$\frac{\partial f}{\partial t} + \sum_{i=1}^{6} \dot{z}_i \frac{\partial f}{\partial z_i} = 0, \tag{48}$$

where $\mathbf{z} = (\mathbf{x}, \mathbf{v})$. But this equation is covariant with respect to arbitrary changes of co-ordinates, and therefore is valid in any co-ordinate system. (There is no restriction to canonical transformations.) For example, in the guiding centre co-ordinates $(\mathbf{X}, U, \Theta, \mu)$, the Vlasov equation is

$$\frac{\partial f}{\partial t} + \dot{\mathbf{X}} . \nabla f + \dot{U} \frac{\partial f}{\partial U} + \dot{\Theta} \frac{\partial f}{\partial \Theta} + \dot{\mu} \frac{\partial f}{\partial \mu} = 0, \tag{49}$$

where $f = f(\mathbf{X}, U, \Theta, \mu, t)$. This conclusion in no way depends on the detailed character of the variables $(\mathbf{X}, U, \Theta, \mu)$, e.g. the fact that they are averaged on

the gyro time-scale. Of course, if the entire plasma should obey the same length and time-scales as go into the guiding centre approximation (gyroradius $\ll$ scale length, etc.), then the last two terms above would vanish, for we would have $\partial f/\partial\Theta = 0$ and $\dot{\mu} = 0$. But if there were a wave superimposed on a quiescent background, and if the guiding centre variables were related to the particle co-ordinates by using only the background fields, according to (30)–(31), then we would have $\partial f/\partial\Theta \neq 0$ and $\dot{\mu} \neq 0$, owing to the effects of the wave. But (49) would still be the correct Vlasov equation in the variables $(\mathbf{X}, U, \Theta, \mu)$. If the Vlasov equation is needed in some arbitrary co-ordinate system $\mathbf{z}$, then one can compute the $\dot{z}$'s by any means, e.g. by using a Lagrangian, and then substitute into (48).

Liouville's theorem is not the same as the Vlasov equation, since conserving phase volume is not the same as conserving particles. (As is well known, the Vlasov equation can be derived from the Liouville equation in $6N$-dimensional phase space, by neglecting correlations. But here we are only referring to the 6-dimensional phase space.) But Liouville's theorem, (41), can be combined with the Vlasov equation to give results which are completely covariant. In particular, we have

$$\frac{\partial F}{\partial t} + \sum_{i=1}^{6} \frac{\partial}{\partial z_i}(F\dot{z}_i) = 0. \tag{50}$$

This equation represents the conservation of particles in its simplest form, and it would be true even for dissipative forces (whereas (48) would not).

For example, if the entire plasma obeys the same length and time-scale requirements that go into the guiding centre approximation, then the Vlasov equation can be written

$$\frac{\partial F}{\partial t} + \nabla.(F\dot{\mathbf{X}}) + \frac{\partial}{\partial U}(F\dot{U}) = 0, \tag{51}$$

where $\dot{\mathbf{X}}$ and $\dot{U}$ are given by (8)–(9), or some higher-order version. Note that if we write $F = Df$, with $D = B_{\parallel}^{*}$, then D exactly cancels the denominators in (8)–(9), and the Vlasov equation takes on the interesting form

$$\frac{\partial}{\partial t}(fB^{*}) + \nabla.\{f[U\mathbf{B}^{*} + \epsilon\hat{b}\times(\mu\nabla B - \mathbf{E}^{*})]\} - \frac{\partial}{\partial U}[f\mathbf{B}^{*}.(\mu\nabla B - \mathbf{E}^{*})] = 0. \tag{52}$$

This result would be even simpler in Boozer's (1970) co-ordinates, using (11).

In the single particle picture, it is actually never necessary to write down the Vlasov equation. Instead, one uses the fact that a function f is a solution to the Vlasov equation if and only if it is a constant of the single particle motion. This fact is well known, but it is often insufficiently exploited. Of course, this fact does not obviate the requirement of determining fields self-consistently.

For example, it is common in practice to use the quantity $\mu_0 = w^2/2B(\mathbf{x}, t)$ as a phase space co-ordinate in manipulating gyrokinetic equations. However, since μ_0 is itself not conserved (it is only the first term of an expansion), it would be better to use the entire μ, shown in (31), as a co-ordinate. This causes an expansion of f in powers of ϵ, $f = f_0 + \epsilon f_1 + \ldots$, to depend on the co-ordinates z

employed, since μ itself is now ordered. But this is a minor annoyance compared with retaining the $O(\epsilon)$ term of (31), or worse yet, rederiving it unnecessarily.

The advantage of using Hamiltonian methods for gyrokinetic equations is that they provide very powerful and efficient ways of determining the constants of single particle motion, especially when nonlinear effects are under consideration. Effectively, this is applying Noether's theorem to self-consistent problems. Nonlinear effects have recently been of interest, as is shown in the work of Wong (1981) and Frieman & Chen (1982). We will report on the use of Hamiltonian methods in such calculations in the future.

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REFERENCES

ARNOLD, V. I. 1978 *Mathematical Methods of Classical Mechanics.* Springer.

BOOZER, A. H. 1980 *Phys. Fluids*, **23**, 904.

CARY, J. R. 1979 Ph.D. Thesis, University of California, Berkeley.

CARY, J. R. 1981 *Phys. Rep.* **79**, 131.

CARY, J. R. & LITTLEJOHN, R. G. 1982 (To be published.)

DRAGT, A. J. & FINN, J. M. 1976 *J. Math. Phys.* **17**, 2215.

FRIEMAN, E. A. & CHEN, L. 1982 *Phys. Fluids*, **25**, 502.

GOLDSTEIN, H. 1980 *Classical Mechanics*, 2nd ed. Addison-Wesley.

HASTIE, R. J., TAYLOR, J. B. & HAAS, F. A. 1967 *Ann. Phys.* **41**, 302.

KRUSKAL, M. D. 1965 *Plasma Physics*, p. 67. IAEA.

LANDAU, L. D. & LIFSHITZ, E. M. 1969 *Mechanics*, 2nd ed. Pergamon.

LITTLEJOHN, R. G. 1979 *J. Math. Phys.* **20**, 2445.

LITTLEJOHN, R. G. 1981 *Phys. Fluids*, **24**, 1730.

LITTLEJOHN, R. G. 1982*a* *J. Math. Phys.* **23**, 742.

LITTLEJOHN, R. G. 1982*b* *Physica Scripta*, (To be published.)

MOROZOV, A. I. & SOLOV'EV, L. S. 1966 *Reviews of Plasma Physics* (ed. M. A. Leontovich), vol. 2, p. 201. Plenum.

NORTHROP, T. G. 1963 *The Adiabatic Motion of Charged Particles.* Interscience.

NORTHROP, T. G. & ROME, J. A. 1978 *Phys. Fluids*, **21**, 384.

TAYLOR, J. B. 1964 *Phys. Fluids*, **7**, 767.

WHITHAM, G. B. 1974 *Linear and Nonlinear Waves.* Interscience.

WONG, H. V. 1981 *University of Texas Fusion Research Center Report* FRCR 230.